

Governed AI Decision Systems: Formalizing Evidence-Policy Consistency and Architectural Separation in Adaptive Enterprise Software

Executive Summary

The rapid integration of Large Language Models and autonomous agentic workflows into enterprise decision-making has exposed an architectural vulnerability: the fundamental tension between stochastic adaptability and deterministic policy compliance. While modern statistical language models demonstrate capabilities in contextual reasoning, unstructured document parsing, and generalized problem-solving, their probabilistic nature prevents them from guaranteeing strict adherence to non-negotiable organizational, legal, and regulatory constraints. Best-effort prompt engineering and simple guardrails frequently fail under edge cases and adversarial pressure, yielding policy compliance pass rates as low as 31% to 69% in agentic enterprise tasks.

This report presents an investigation into the design, formalization, and engineering of Governed AI Decision Systems. It establishes that while individual sub-problems—such as access-control policy enforcement, post-hoc model explainability, or threshold-based human escalation—have been independently addressed in academic literature, their integration into a unified, domain-agnostic enterprise software system remains an unmapped research frontier. The primary research contribution formalized herein is the Evidence-Policy Consistency (EPC) problem: algorithmically verifying whether the evidence extracted by an adaptive AI system is sufficient, non-contradictory, and legally or procedurally admissible under the deterministic policies governing a decision domain.

Furthermore, this report bridges theoretical formalization with practical software engineering by defining the Governed AI Decision Pipeline (GADP) architecture. This architecture operationalizes an explicit architectural separation of powers between evidence extraction, policy adjudication, and decision execution. Using a career automation ecosystem—incorporating candidate resume analysis, automated job matching, and adaptive career mentorship—as an initial enterprise benchmark, this report details the concrete database schemas, multi-signal uncertainty engines, and execution flows necessary to build production-grade, policy-governed AI software.

Taxonomy of Related Research Domains

To position Governed AI Decision Systems within the computer science landscape, the underlying research ecosystem must be structured into three analytical tiers based on functional proximity, structural overlap, and application domains.

Tier 1: Directly Adjacent Research Domains

Tier 1 encompasses the foundational technical sub-disciplines that directly inform the mechanisms of policy enforcement, evidence evaluation, and system accountability. AI Policy Enforcement focuses on mechanisms that translate written governance rules into dynamic, real-time computational controls to block, permit, or log agent actions. Runtime Verification for LLM Agents involves dynamic monitoring of agent trajectory outputs against formal properties without requiring model re-training. Governance-as-a-Service (GaaS) represents externalized, decoupled software layers that intercept multi-agent communications to enforce behavioral sanctions and trust boundaries. Human-in-the-Loop (HITL) Decision Support encompasses systems that route automated tasks to human experts based on decision boundaries, uncertainty thresholds, or risk profiles. Decision Provenance and Audit Trails provide sourcing methodologies that capture data lineage, model versions, evidence context, and execution steps to create immutable records. Explainable AI (XAI) focuses on post-hoc feature attribution techniques designed to expose internal model weights and input feature importance. Uncertainty Quantification (UQ) for LLMs supplies mathematical methods measuring model epistemic (knowledge-based) and aleatoric (data-inherent) uncertainty to derive confidence metrics.

Tier 2: Structural Overlap Domains

Tier 2 represents adjacent systems and governance fields whose architectural patterns overlap with enterprise AI decision frameworks. Multi-Agent Systems Governance establishes structural rules governing interaction protocols, contract negotiation, and consensus mechanisms across distributed autonomous agents. Enterprise AI Workflow Automation involves the industrial orchestration of multi-step business logic combining deterministic enterprise resource planning flows with probabilistic models. Formal Verification and Program Synthesis supplies theoretical methods proving that a system's execution strictly adheres to mathematical specifications. Regulatory AI and Compliance Frameworks encompass computational paradigms designed to operationalize legal mandates such as the EU AI Act, NIST AI Risk Management Framework (RMF), and GDPR Article 22. Knowledge Graphs and Governed Retrieval-Augmented Generation (RAG) provide graph-structured evidence retrieval mechanisms enforcing semantic relationships and access permissioning over data sources prior to model generation.

Tier 3: Instantiation Domains

Tier 3 defines high-stakes enterprise application domains where non-compliance carries severe financial, legal, or societal penalties, demanding formal policy governance. Human Resources and Recruitment Systems involve automated candidate screening, skill matrix evaluation, and Applicant Tracking Systems (ATS) governed by Equal Employment Opportunity Commission (EEOC) mandates. Financial Underwriting and

Credit Decisions utilize automated loan, credit, and insurance approval pipelines bounded by fair lending regulations. Clinical Decision Support Systems implement diagnostic and treatment recommendation engines operating under HIPAA regulations and medical evidence standards. Legal Compliance and Regulatory Review applications deploy enterprise document processing platforms determining regulatory exposure, contract admissibility, and policy violations.

Domain Tier	Research Field	Core Operational Mechanism	Primary State-of-the-Art Limitations
Tier 1	Runtime Verification	Dynamic constraint monitoring of agent action trajectories.	Strictly deterministic; lacks probabilistic policy reasoning.
Tier 1	Decision Provenance	Source-to-decision audit logging and evidence lineage.	Domain-siloed; conflates internal XAI with evidence origin.
Tier 1	Uncertainty Quantification	Model confidence scoring via output entropy and variance.	Poorly calibrated in LLMs; confuses epistemic with aleatoric noise.
Tier 2	Governed RAG	Constraint-filtered document retrieval for generative prompts.	Filters for semantic relevance rather than legal policy admissibility.
Tier 2	Formal Verification	Algorithmic proof generation for code and logic pipelines.	High computational complexity; struggles with natural language rules.
Tier 3	HR / Recruitment AI	Structured matching of candidate profiles to job specifications.	Over-reliance on opaque ATS scores; vulnerable to demographic bias.

Literature Mapping and Comparative Architecture Analysis

An analysis of literature from 2023 through 2026 reveals a fragmented research landscape. Existing frameworks address localized concerns—such as dialogue safety, API interception, or clinical data lineage—but lack an integrated architecture that simultaneously guarantees policy determinism, context sensitivity, and evidence admissibility.

In deterministic enforcement research, AgentSpec (ICSE 2026) defines a domain-specific specification language for formalizing runtime behavioral controls over agent tool access and action sequences. While it successfully guarantees state transition boundaries, its capabilities are strictly deterministic. It cannot evaluate ambiguous

natural language policy conditions or adapt to context-dependent policy exemptions without manual rule re-writing. Similarly, integrating Open Policy Agent (OPA) with MLflow establishes a strict operational boundary by executing policy evaluation in a deterministic Rego engine, using the LLM solely as an unstructured feature extractor. While this guarantees zero policy violation false-positives for hard rules, it collapses when policies contain natural language ambiguity, forcing human developers to manually codify complex real-world logic into rigid boolean expressions.

In generative and multi-agent governance, Governance-as-a-Service (GaaS - arXiv:2508.18765) proposes an externalized multi-agent architecture that intercepts agent communications to enforce behavioral sanctions based on accumulated trust factors. GaaS excels at preventing unauthorized API calls and system-level actions. However, it operates as an action-level firewall; it remains blind to evidence quality, failing to verify whether the data used by an agent to justify a decision meets organizational admissibility standards.

In natural language policy translation, Policy-as-Prompt (arXiv:2509.23994) converts unstructured policy documents into dynamic LLM guardrail prompts. While this approach bridges the policy-to-code translation gap, it relies on the LLM itself to enforce the translated constraints. This design re-introduces non-deterministic reasoning into the compliance pipeline, yielding high pass rate variations and susceptibility to prompt injection vulnerabilities. Similarly, PolicyGuard (arXiv:2606.29225) deploys dialogue-grounded sub-agents for verifier-based policy adherence, but suffers from supervisory blind spots where verifier agents inherit the miscalibrations of the models they supervise .

In clinical provenance and routing, Clinical AI Provenance Frameworks (PMC 2026) demonstrate robust architectures for tracking source data lineage, model versions, and immutable audit trails. However, these architectures are deeply domain-tied to medical diagnostics and lack connection to dynamic policy enforcement engines or generalized confidence-based human routing protocols. On the routing side, systems like CascadeDebate (ACL 2026) and HULA formalize escalation through uncertainty-aware multi-agent deliberation and return variance estimation, yet remain vulnerable to the underlying miscalibration of generative confidence scores.

Framework	Primary Evaluation Paradigm	Context Sensitivity	Audit Trail Completeness
AgentSpec [cite: 1]	Deterministic Rule Check	Low	Medium

Framework	Primary Evaluation Paradigm	Context Sensitivity	Audit Trail Completeness
NeMo Guardrails [cite: 1]	Dialogue Scripting (Colang)	Medium	Low
GaaS [cite: 1]	Trust-Factor Pattern Check	Low	High (In Logs)
Policy-as-Prompt [cite: 1]	Probabilistic LLM Evaluation	High	Low
OPA + MLflow [cite: 1]	Static Rule Adjudication	Low	High (Re Logs)
Clinical AFPR [cite: 1]	Immutable Provenance RAG	High	High (Lit Tracking)
CascadeDebate [cite: 1]	Multi-Agent Deliberation	High	Low

Cross-Domain Problem Matrix

Analyzing systemic operational failures across enterprise domains reveals recurring patterns where probabilistic AI reasoning clashes with deterministic organizational requirements.

Problem Identifier	Affected Domains	Current Industry Interventions	Structural Failure Vulnerabilities
Policy-to-Practice Gap	Recruitment, Financial Lending, Legal Review	Policy-as-Prompt, System Prompt Guardrails.	System prompt or complex edge compliance rat
Decision Provenance Sinks	HR Screening, Clinical Support, Credit Scoring	Post-hoc XAI (SHAP/LIME), Feature Importance Logs.	Feature weight data lineage. T evidence was t
Uncalibrated Routing	Enterprise Customer Support, ATS Screening	Static LLM confidence score thresholds.	LLMs exhibit se outputs. Single

Problem Identifier	Affected Domains	Current Industry Interventions	Structural Failure Vulnerabilities
			auto-approvals
Determinism vs. Context	Security Operations, Infrastructure Agents	Hardcoded Open Policy Agent (OPA) rules.	Deterministic enforcement due to a lack of administrative flexibility
Adversarial Policy Bypass	Autonomous Agents, ATS Resume Parsers	Prompt Sanitization, Input Filtering Layers.	Indirect prompt injection through unstructured data filters.
Policy Staleness	Regulatory Compliance, Healthcare Administration	Quarterly manual policy refactoring.	Codebases lag resulting in stale policy transition windows
Epistemic Ambiguity Blindness	Clinical AI, HR Candidate Evaluation	Global variance/entropy thresholds.	Systems conflate uncertainty (epistemic) with misrouting hallucinations

Literature Gap Analysis and Problem Formalization

The primary literature gaps center around five structural deficiencies: the policy-to-practice gap in unstructured environments where systems cannot reliably convert natural language rules into execution constraints while preserving context; the lack of formal evidence governance, as current architectures treat retrieval relevance as synonymous with policy admissibility; the conflation of explainability and provenance, where internal feature attributions are used in place of verifiable data lineage; inadequate uncertainty routing that relies on uncalibrated self-confidence scores without separating missing knowledge from input noise; and the complete absence of an integrated architecture combining policy-aware evidence collection, out-of-model deterministic enforcement, multi-signal uncertainty quantification, and immutable provenance logging.

To resolve these literature gaps, the Evidence-Policy Consistency (EPC) problem is formalized as a dynamic verification problem. Let an organizational policy P be defined as a structured set of m discrete rules:

$$P = \{r_1, r_2, \dots, r_m\}$$

Each rule $r_j \in P$ is composed of a pair (h_j, s_j) , where h_j represents a deterministic hard constraint (a non-negotiable legal or enterprise boundary), and s_j represents a soft

contextual evaluation criterion (a natural language requirement requiring semantic interpretation).

Let D represent an unstructured input document corpus (such as an applicant resume, job description, or career profile). An adaptive AI extraction engine E processes D under execution context C (containing environmental parameters, user roles, and jurisdictional constraints) to extract a structured evidence set E :

$$E(D, C) \rightarrow E = \{e_1, e_2, \dots, e_n\}$$

An individual evidence item $e_i \in E$ is evaluated against an admissibility predicate $A(e_i, r_j, C) \in \{0, 1\}$, which verifies source lineage, temporal freshness, and regulatory compliance. The subset of policy-admissible evidence mapped to rule r_j is defined as:

$$E_{r_j} = \{e_i \in E \mid A(e_i, r_j, C) = 1\}$$

The Evidence-Policy Consistency function $F_{EPC}(E, P, C)$ evaluates whether E provides a valid, sufficient, and non-contradictory foundation for executing automated decision $d \in D$:

$$F_{EPC}(E, P, C) = \bigwedge_{j=1}^m [(\forall e_i \in E_{r_j}, A(e_i, r_j, C) = 1) \wedge S(E_{r_j}, h_j) \wedge \neg C(E_{r_j})]$$

In this formulation, $S(E_{r_j}, h_j) \in \{0, 1\}$ is a deterministic predicate evaluating whether evidence set E_{r_j} completely satisfies hard constraint h_j . The term $C(E_{r_j}) \in \{0, 1\}$ is a boolean function returning 1 if internal contradictions exist within evidence subset E_{r_j} .

If $F_{EPC}(E, P, C) = 1$, the system emits a compliance certificate $K = \langle d, E, P, C, F_{EPC}, \sigma \rangle$ secured by a cryptographic signature σ , authorizing decision execution. If $F_{EPC} = 0$, automated decision execution is blocked, and the system triggers confidence-aware human escalation.

Analytical Deconstruction of Core Assumptions

Designing a practical enterprise decision system requires challenging eight foundational assumptions common in AI governance literature.

Regarding the role of LLMs in policy decisions, the literature often assumes that generative models should directly execute policy evaluation and decide whether an enterprise action is compliant. Statistical models, however, cannot guarantee zero-shot compliance due to output stochasticity. A refined architectural positioning mandates that LLMs must be strictly relegated to unstructured evidence extraction, semantic translation, and contextual parsing, while final policy adjudication must be executed out-of-model by deterministic boolean rule engines.

Concerning policy determinism, prevailing frameworks treat enterprise policies as rules that can be fully converted into static code. Real-world enterprise policies contain intentional natural language ambiguity to allow human managerial discretion. Policies must therefore be architected as a bifurcated structure: a deterministic hard core representing bright-line constraints surrounded by an LLM-assisted contextual periphery for soft judgment criteria.

Regarding AI output paradigms, autonomous agent designs frequently assume that systems should produce final, direct decisions across all operational scenarios. Direct execution in regulated domains creates unacceptable liability exposure under legal regimes like GDPR Article 22. High-risk AI tasks should instead produce structured, policy-admissible evidence packages that feed downstream decision engines, reserving direct decision generation for high-confidence, low-risk scenarios.

In human-in-the-loop routing, reliance on unimodal LLM self-reported confidence scores presents a failure point. Generative models exhibit systematic overconfidence, frequently hallucinating false justifications with high self-reported probabilities. Escalation routing must combine multiple uncertainty signals—including model output entropy, evidence coverage metrics, policy ambiguity scores, and historical failure distributions.

Regarding evidence governance versus explainability, feature attribution maps (such as SHAP or LIME) are often assumed to satisfy regulatory demands for system transparency. Explainable AI methods explain internal model mechanics, revealing which token representations influenced an output, but fail to prove whether external source documents were untampered, legally obtained, or procedurally admissible. Decision provenance—tracking external document lineage and immutable snapshot logging—must be treated as an independent architectural layer distinct from model interpretability.

On the relationship between provenance and stakeholder trust, exposing comprehensive end-to-end data provenance logs is frequently assumed to directly increase user trust. Raw provenance dumps overload human reviewers with complex technical metadata, degrading decision speed and increasing cognitive fatigue.

Systems must provide role-differentiated visibility: concise executive summaries for operational reviewers, paired with cryptographically verifiable audit ledgers for compliance officers.

Regarding routing triggers in human-in-the-loop frameworks, standard designs trigger human review whenever confidence falls below a global threshold. Low confidence caused by aleatoric uncertainty (such as inherent document noise or formatting typos) cannot be resolved by human intervention any better than by automated systems, wasting human capital. Human review must be selectively triggered by epistemic uncertainty (missing knowledge or uncovered policy rules) and policy ambiguity, filtering out aleatoric noise.

Finally, regarding the locus of policy enforcement, compliance is frequently sought via model alignment methods (such as RLHF or system prompt tuning). In-weights alignment is inherently probabilistic and susceptible to jailbreaking, prompt injection, and adversarial distribution shifts. Policy enforcement must exist as an external, out-of-model middleware layer that intercepts model outputs prior to execution.

Comparative Analysis of Frameworks and Open-Source Ecosystems

A systematic structural assessment highlights the functional gaps in existing governance implementations.

Architectural Feature	Open Policy Agent (OPA)	NeMo Guardrails	Governance-as-a-Service (GaaS)	AgentSpec	Clinic Fram
Deterministic Hard Enforcement	Native (Rego Engine)	Partial (Colang Scripts)	Native (Pattern Rules)	Native (DSL Trajectories)	None Only)
Contextual Policy Interpretation	None (Static Inputs)	Medium (LLM Dialog)	Low (Action Matching)	None (State Machine)	High (LLMs
Evidence Admissibility Checking	None	None	None	None	Partial Rules
Multi-Signal Uncertainty Routing	None	None	None	None	None
Tamper-Evident Provenance	External	External	Audit Logs	Execution Logs	Native (Cryp

Architectural Feature	Open Policy Agent (OPA)	NeMo Guardrails	Governance- as-a-Service (GaaS)	AgentSpec	Clinic Frame
Logging					Ledge
Domain Agnosticism	High	Low	Medium	High	Low (Speci

Research Question Taxonomy and Frontier Discovery

To systematically address the technical gaps in Governed AI Decision Systems, potential research questions are mapped across four operational difficulty tiers.

Difficulty Tier	ID	Research Question Formulations
Beginner (Engineering)	RQ1	How can Open Policy Agent (OPA) schemas be automatically compiled from unstructured HR policy documents?
Beginner (Engineering)	RQ2	What JSONB database schema optimizations preserve query performance when logging full LLM decision lineage?
Beginner (Engineering)	RQ3	How do candidate UI layouts for provenance visualization impact human reviewer decision latency?
Intermediate (Systems)	RQ4	How can an out-of-model policy interceptor remain model-agnostic while preserving structured context across stateful agent chains?
Intermediate (Systems)	RQ5	What formal specification DSL best captures compliance constraints in natural language employment policies?
Intermediate (Systems)	RQ6	How can evidence provenance logs be selectively redacted for GDPR compliance without invalidating cryptographic audit trails?
Advanced (Theoretical)	RQ7	Policy-Admissible Evidence Formalization: What formal logic can determine whether retrieved evidence satisfies the admissibility standards of a governing policy?
Advanced (Theoretical)	RQ8	Epistemic vs. Aleatoric Uncertainty Routing: Can multi-modal uncertainty decomposition reliably separate missing evidence from text ambiguity?
Advanced (Theoretical)	RQ9	Compositional Policy Enforcement: How do deterministic compliance guarantees compose when autonomous agents with distinct policy constraints interact?

Difficulty Tier	ID	Research Question Formulations
Frontier (Publication)	RQ10	The Evidence-Policy Consistency (EPC) Problem: Algorithmical verifying evidence sufficiency, admissibility, and non-contradiction under unstructured constraints.
Frontier (Publication)	RQ11	Governed Uncertainty Propagation: Frameworks for propagating uncertainty bounds across multi-step agent pipelines with mathematical false-negative guarantees.
Frontier (Publication)	RQ12	Policy-Aware Evidence Selection in RAG: Optimization algorithm selecting retrieval contexts based on legal admissibility rather than semantic similarity.
Frontier (Publication)	RQ13	Architectural Separation of Powers in Agentic AI: Formal proofs establishing that decoupling evidence collection, policy evaluation, execution is necessary for compliance.
Frontier (Publication)	RQ14	GADS-Bench: The first standardized benchmark evaluating agentic compliance under active, adversarial edge-case conditions.

Ranked Research Opportunities

Evaluating research avenues requires analyzing trade-offs across novelty, practical impact, academic rigor, feasibility, and publication potential.

Rank	Research Direction	Novelty	Practical Impact	Academic Impact
1	Evidence-Policy Consistency Formalization	High	High	High
2	Governed AI Architecture (Separation of Powers)	High	High	High
3	Policy-Aware Evidence Selection in RAG	Medium-High	High	High
4	Governed Uncertainty Propagation	High	Medium	High
5	Benchmark for Governed Decision Systems (GADS-Bench)	Medium	Very High	High
6	Epistemic vs. Aleatoric Uncertainty Routing	Medium	High	Medium

Rank	Research Direction	Novelty	Practical Impact	Academic Impact
7	Compositional Multi-Agent Policy Enforcement	High	Medium	High
8	Dynamic Trust Boundary Specification	Medium	High	Medium

The formalization of Evidence-Policy Consistency (Rank 1) represents the most viable academic and practical contribution. The problem sits at the intersection of formal verification, legal reasoning, and statistical AI—three fields that currently lack unifying mathematical frameworks. It addresses immediate regulatory pressures (such as the EU AI Act and EEOC compliance mandates) while providing a rigorous theoretical foundation suitable for top-tier publication.

Production System Blueprint and Enterprise Implementation Strategy

To validate the theoretical contributions of the Evidence-Policy Consistency framework, the Governed AI Decision Platform (GADP) is implemented as an enterprise-grade backend architecture. The target domain instantiation selected for this blueprint is a candidate resume screening, ATS optimization, and career mentoring system. This domain features complex policy requirements, including EEOC non-discrimination mandates, mandatory technical skill prerequisites, and high-stakes automated decisions.

System Architecture Overview

The system is constructed using FastAPI, Python 3.14+, PostgreSQL with native JSONB, SQLAlchemy 2.x ORM, Alembic migrations, and the Google Gemini API.

The architectural flow begins at the API Layer, where request routers constructed in FastAPI handle incoming data streams. Unstructured artifacts—such as candidate resumes, job descriptions, or career goal profiles—enter the Unstructured Ingestion Pipeline. Plain text is extracted alongside parsed structural metadata and normalized into standard formats.

Next, the extracted data passes to the Evidence Extraction and Admissibility Layer. Here, the Google Gemini API functions as an LLM context parser, extracting structured candidate claims and skill matrices into Pydantic v2 schemas. Before these claims are used downstream, an Admissibility Evaluator verifies source lineage, document freshness, and regulatory admissibility.

Once admissible evidence is mapped, control passes out of the LLM to the Out-of-Model Deterministic Policy Engine. Hard constraints (such as required degrees or certifications) are dispatched to an Open Policy Agent (OPA) Rego evaluation engine, while soft contextual criteria are evaluated via constrained semantic interpreters.

In parallel, the Multi-Signal Uncertainty Quantification Engine calculates epistemic coverage gaps, aleatoric source variance, and policy ambiguity scores to yield a composite routing metric. If the composite score falls below the escalation threshold and all hard constraints are satisfied, the system executes Automated Decision Execution, emitting a signed compliance certificate. If uncertainty exceeds the threshold or a hard rule is violated, the transaction is diverted to the Human-in-the-Loop Escalation Router. Finally, every execution step, source document snapshot, and evaluation result is committed to the Immutable Audit Ledger as JSONB compliance snapshots.

Database Schema Architecture

The database architecture extends core business tables (`User` , `Resume` , `ResumeVersion` , `Job` , `Application` , `SkillRoadmap`) with dedicated AI governance entities (`PolicyDocument` , `EvidenceLedger` , `ComplianceCertificate` , `AuditTrailLog`). This separation isolates business operational states from immutable AI decision records.

In terms of entity relationships, a `User` owns multiple `Resume` instances and `Application` records. Each `Resume` tracks history through one-to-many immutable `ResumeVersion` snapshots. A `ResumeVersion` maintains one-to-one linkage with a `ResumeAnalysis` record, and one-to-many relationships with `CompatibilityReport` and `CoverLetter` entities, which both reference a target `Job` . The governance layer attaches directly to `Job` through one-to-many `PolicyDocument` entities, which govern `EvidenceLedger` instances linked to `ResumeVersion` snapshots. Each `EvidenceLedger` generates a unique `ComplianceCertificate` , which anchors the `AuditTrailLog` history.

Entity Name	Primary Key	Foreign Keys / Index Keys	Key Schema Attribute (PostgreSQL Data Type)
User	id (Int)	provider_id (Unique, String)	email (String), created_at (DateTime)
ResumeVersion	id (Int)	resume_id (FK -> Resume.id)	storage_key (String), raw_text (Text), parsed_resume (JSONB)

Entity Name	Primary Key	Foreign Keys / Index Keys	Key Schema Attribute (PostgreSQL Data Type)
Job	id (Int)	company (String), title (String)	description (Text), parsed_requirements (JSONB)
PolicyDocument	id (Int)	job_id (FK -> Job.id)	version (String), hard_constraints (JSONB), soft_criteria (JSONB)
EvidenceLedger	id (Int)	resume_version_id (FK), policy_id (FK)	extracted_evidence (Text), admissibility_map (JSONB)
CompatibilityReport	id (Int)	resume_version_id (FK), job_id (FK)	compatibility_score (Float), matched_skills (JSONB), missing_skills (JSONB)
ComplianceCertificate	id (Int)	evidence_ledger_id (FK, Unique)	is_compliant (Boolean), uncertainty_vector (JSONB), signature (String)
AuditTrailLog	id (Int)	certificate_id (FK), user_id (FK)	execution_trace (Text), timestamp (DateTime)

End-to-End Execution Lifecycle

The runtime decision lifecycle follows a six-stage pipeline.

In Stage 1 (Document Upload and Raw Artifact Storage), the user uploads an unstructured candidate resume (PDF or DOCX format) via the FastAPI upload router. The file is stored in object storage, producing a unique `storage_key`. Extracted plain text is persisted alongside an immutable `ResumeVersion` database record.

In Stage 2 (Schema Parsing and Evidence Extraction), the `ResumeVersion` text is dispatched to the Gemini API using Pydantic target schemas. The model parses unstructured text into structured JSONB representations covering candidate skills, work experience, education metrics, and accomplishments. The result is written directly to `ResumeVersion.parsed_resume`.

In Stage 3 (Policy Compilation and Admissibility Evaluation), the system retrieves the `PolicyDocument` associated with the target `Job`. Policy rules are split into

deterministic hard constraints (h_j) and contextual soft criteria (s_j). An `EvidenceLedger` entry is initialized, mapping extracted resume claims to corresponding policy rules while evaluating admissibility predicates (A) based on document lineage and verification metadata.

In Stage 4 (Deterministic Policy Evaluation), hard constraints (h_j) are evaluated out-of-model by the OPA Rego engine. If a candidate fails a mandatory requirement (such as lacking a required certification), $S(E_{r_j}, h_j)$ evaluates to 0. The engine immediately flags a deterministic policy violation, blocking automated application processing regardless of candidate compatibility scores.

In Stage 5 (Multi-Signal Uncertainty Quantification and Routing), for candidates clearing hard constraints, soft criteria (s_j) are evaluated to determine overall job compatibility. The UQ Engine calculates model output variance, evidence coverage ratios, and policy ambiguity metrics. If the composite uncertainty score $U_{comp} \leq \tau_{route}$ and no hard violations exist, the system issues a `CompatibilityReport`, generates a cryptographically signed `ComplianceCertificate`, and authorizes automated application status updates. If $U_{comp} > \tau_{route}$ or evidence contradictions ($C(E_{r_j}) = 1$) are detected, automated execution is suspended, creating a `RoutingLog` that routes the application to human reviewers alongside a visual summary of missing coverage areas.

In Stage 6 (Audit Snapshotting), all prompt contexts, parsed JSON payloads, evidence mappings, rule evaluation logs, and uncertainty vectors are committed to an immutable `AuditTrailLog` entry linked via PostgreSQL JSONB fields. The certificate is cryptographically signed using the system's private key to guarantee non-repudiation.

Academic Research Roadmap and White Paper Strategy

Establishing Governed AI Decision Systems as a recognized research sub-field requires pairing theoretical formalization with structured benchmark contributions.

Recommended White Paper Specification

The white paper should be titled "Governed AI Decision Systems: Formalizing Evidence-Policy Consistency and Architectural Separation in Adaptive Automated Decision-Making". The core thesis establishes that current AI governance approaches incorrectly treat runtime policy enforcement and evidence management as separate concerns. True policy compliance requires verifying both the deterministic execution of hard constraints and the evidentiary admissibility of data used for contextual judgments. Decoupling evidence collection, policy evaluation, and decision execution is necessary to achieve verifiable regulatory compliance.

Target publication outlets for this work include top-tier systems and AI venues. The primary systems target is the IEEE Symposium on Security and Privacy (S&P) or the ACM International Conference on Software Engineering (ICSE). The primary AI theory target is the AAAI Conference on Artificial Intelligence or the NeurIPS Datasets and Benchmarks Track.

Three-Year Execution Roadmap

The roadmap organizes technical milestones across three years, balancing theoretical formalization, benchmark creation, and production software implementation.

Year & Focus	Academic & Technical Milestones	Operational De
Year 1: Foundations & Prototyping	Q1-Q2: Theoretical survey and mathematical formalization of EPC definitions.	Comprehensive li problem specifica
	Q3: White paper publication and core backend prototype development.	Release white pa on FastAPI and F
	Q4: Single-domain validation on recruitment ATS workloads.	Benchmark GADI screeners using c
Year 2: Benchmarking & Theory	Q1-Q2: GADS-Bench benchmark design and data collection.	Publish first benc compliance unde
	Q3: Training and evaluation of Evidence-Policy Consistency verifiers.	Publish standalor evidence admissi
	Q4: Development of policy-aware governed RAG modules.	Extend retrieval n legal admissibility
Year 3: Multi-Agent & Platform	Q1: Formal verification proofs for multi-agent policy composition.	Publish compositi agent networks.
	Q2: Production open-source release of the GADP platform.	Release core eng dashboards.
	Q3: Regulatory alignment mapping to the EU AI Act and NIST RMF.	Publish formal co enterprise regulat
	Q4: Domain expansion into financial lending and clinical support.	Extend benchmai financial underwri

Strategic Synthesis and Conclusion

The theoretical and architectural analyses presented in this report demonstrate that guaranteeing enterprise policy compliance in adaptive AI decision systems is an

achievable engineering reality. The primary research challenge is not the elimination of probabilistic AI models, but rather the construction of software architectures that properly constrain them.

By formalizing the Evidence-Policy Consistency (EPC) problem, researchers obtain a rigorous foundation for evaluating whether AI decisions are grounded in sufficient, non-contradictory, and legally admissible source evidence. By operationalizing the Governed AI Decision Platform (GADP), software architects gain a concrete model for separating stochastic evidence extraction from deterministic policy adjudication.

When implemented using decoupled backend architectures—supported by immutable database snapshotting, multi-signal uncertainty quantification, and out-of-model policy runtime interceptors—this approach resolves the structural trade-off between AI adaptability and enterprise policy determinism. This framework provides a path toward autonomous decision systems that remain contextually intelligent while remaining verifiable, auditable, and strictly compliant with organizational policies.